Lecture 9

Saturday, 12 September 2026 11:20 AM

EXPECTATION of a RV

$X \to$ discrete $\sim P_X \to \mathbb{E}[X] = \sum_{k \in S_X} k \cdot P_X(k)$

$\searrow$ Conts $\to f_X \to \mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x)\,dx$

$Y = g(X)$. $\mathbb{E}[g(X)] \to \sum_{k \in S_X} g(k) \cdot P_X(k)$

$\hookrightarrow \int_{\mathbb{R}} g(x) f_X(x)\,dx$

Eg - Linear Regression Model

$$y = \omega x$$

	x	y
1	1	0
2	2	5
3	3	6
4	4	9

(x_i, y_i) $l_i(\omega) = \frac{1}{2}(y_i - \omega x_i)^2$

Squared loss $\to \sum_{i=1}^{4} \frac{1}{2}(y_i - \omega x_i)^2$

$\tilde{\omega} = 2$

$$g_i = \frac{\partial}{\partial \omega} l_i(\omega)\Big|_{\tilde{\omega}=2}$$

$\to g_i = -x_i(y_i - \omega x_i)\Big|_{\tilde{\omega}=2} = x_i(2x_i - y_i)$

	x_i	y_i	g_i	
1	1	0	2	✓ *
2	2	5	-2	✓ *
3	3	6	0	✓ *
4	4	9	-4	✓ *

$2(2 \times 2 - 5)$

Evaluate gradient over the dataset

$\frac{1}{4}(\quad + \quad + \quad + \quad)$

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Gradient descent → $\tilde{w} = \tilde{w} - \eta \cdot g_{full}$

$g_{full} = \frac{1}{4}(g_1 + g_2 + g_3 + g_4)$

$= -1$ ✓

Minibatch from the dataset

Select one datapoint at random.

→ G = gradient corresponding to my selected datapoint.

$G \rightarrow$ 2 (↓ 1/4), −2 (↓ 1/4), 0 (↓ 1/4), −4 (↓ 1/4)

$E(G) = \frac{1}{4}(2 - 2 + 0 - 4) = -1$ = g_{full}

Select 2 points at random, with replacement

G_1, G_2 $\quad \bar{G} = \frac{G_1 + G_2}{2}$ ✓

$\bar{G} =$

G_1 \ G_2	2	0	−2	−4
−2	2	1	0	−1
0	1	0	−1	−2
−2	0	−1	−2	−3
−4	−1	−2	−3	−4

$\frac{G_i + G_j}{2}$

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$G \to \{-4, -3, -2, -1, 0, 1, 2\}$

$P(G=g) \to \frac{1}{16}, \frac{2}{16}, \frac{3}{16}, \frac{4}{16}, \frac{3}{16}, \frac{2}{16}, \frac{1}{16}$

$$E[G] = -\frac{4}{16} - \frac{3\times2}{16} - \frac{2\times3}{16} - \frac{4}{16} + \frac{0\times3}{16} + \frac{1\times2}{16} + \frac{2\times1}{16} = -\frac{16}{16}$$

$$= -1 \to g_{\text{fair}}$$

Variance → 'spread' of a dist. around its expectation / mean

$Var(X) \geq 0$

$\sqrt{Var(X)}$ → std. dev.

$$X, \quad Var(X) = E\left((X - \mu_X)^2\right)$$

$$\mu_X \triangleq E(X)$$

$(X - \mu_X)^2$ → sqd. dev. of X from the mean

① $Var(X) \geq 0$

② $Var(X) = E(X^2) - \mu_X^2$

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$$\mathbb{E}\left((X-\mu_X)^2\right) = \mathbb{E}\left(X^2 + \mu_X^2 - 2X\cdot\mu_X\right)$$

$$= \mathbb{E}(X^2) + \mathbb{E}(\mu_X^2) - 2\,\mathbb{E}(X\cdot\mu_X)$$

$$= \mathbb{E}(X^2) + \mu_X^2 - 2\mu_X\,\mathbb{E}(X) \qquad (\mathbb{E}(X) = \mu_X)$$

$$= \mathbb{E}(X^2) - \mu_X^2$$

① $X \sim Ber(p) \rightarrow X = 1$ w.p p, $X = 0$ w.p $1-p$

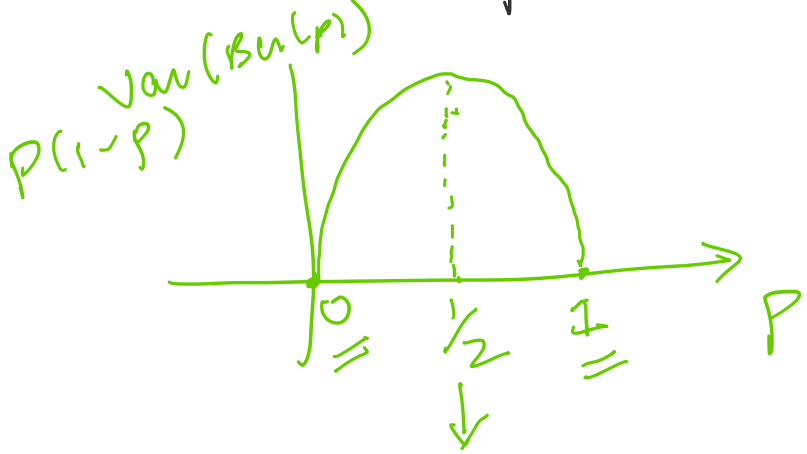
$\mathbb{E}(X) = \mu_X = p$

$Var(X) = \mathbb{E}(X^2) - \mu_X^2$

$= \mathbb{E}(X^2) - p^2 = p - p^2 = p(1-p)$

$\mathbb{E}(X^2) = \sum_{k \in \mathcal{E}_X} g(k)\cdot P_X(k) = 1^2\cdot p + 0^2\cdot(1-p)$ ($g(X) = X^2$)

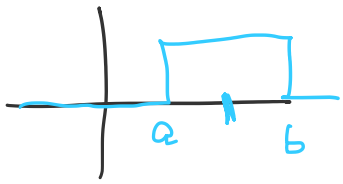
$= 1\cdot p + 0\cdot(1-p) = p$



Eg) $U \sim Unif(a, b)$

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$$f_u(x) = \begin{cases} \frac{1}{b-a} & a < x < b \\ 0 & 0.w \end{cases}$$



$$Var(u) = E(u^2) - E^2(u)$$

$$E(u) = \frac{a+b}{2} \rightarrow \left(\frac{a+b}{2}\right)^2$$

$$E(u^2) = \int_a^b x^2 \cdot f_u(x)\,dx$$

$$= \frac{1}{b-a} \int_a^b x^2\,dx = \frac{1}{b-a}\left[\frac{x^3}{3}\right]_a^b$$

$$= \frac{b^3 - a^3}{3(b-a)}$$

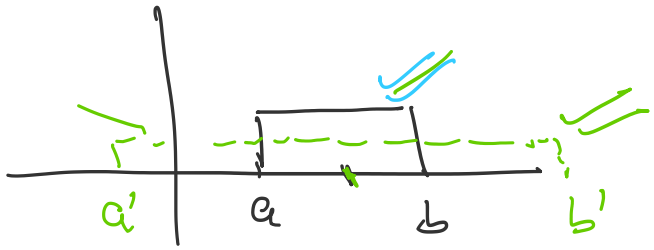
$$Var(u) = \frac{(b-a)^2}{12}$$

Gaussian

$X \sim N(\mu, \sigma^2)$

$\rightarrow E(X)$

$\rightarrow Var(X)$



$$\frac{a+b}{2} = \frac{a'+b'}{2}$$

Score of students $\sim N(10, 1)$

$X \in (\mu_X - \alpha \cdot std(x),\ \mu_X + \alpha \cdot std(x))$?

12 → 9 → 15

$$\Rightarrow \frac{1}{N}\sum (X_i - 10)^2 \sim 1.$$

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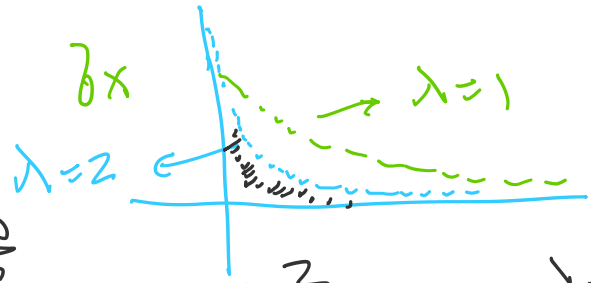
N i.i.d.

$X \sim Exp(\lambda)$ — $Var(X) = ??$

$\lambda \cdot e^{-\lambda x}$

$\frac{1}{\lambda}$ or $\frac{1}{\lambda^2}$

Var ↓ as λ ↑



$$Var(X) = \int_0^\infty (x - \underbrace{\frac{1}{\lambda}}_{\mu_X})^2 \cdot \lambda e^{-\lambda x} \, dx$$

$f_X(x)$

$$Var(X) = \mathbb{E}\left((X - \mu_X)^2\right) = \int_{\mathbb{R}} g(x) \cdot f_X(x) \, dx$$

$g(X)$

$= \mathbb{E}(X^2) - \mu_X^2$

$$= \int_{\mathbb{R}} (x - \mu_X)^2 \cdot f_X(x) \, dx$$

Minibatch

x_i	y_i	g_i
		2
		−2
		0
		−4

→ 1 random sample → G. $\mathbb{E}[G] = g_{full} = -1$

$Var(G) = \mathbb{E}[G^2] - \mathbb{E}^2[G]$

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$G \to \{2, -2, 0, -4\}$ each w.p $1/4$

$E[G^2] = 4\times\frac{1}{4} + 4\times\frac{1}{4} + 0\times\frac{1}{4} + 16\times\frac{1}{4}$

$= 6$

$Var(G) = 6 - 1 = 5$

2 Random samples

$G_1, G_2 \to \frac{G_1 + G_2}{2} = \bar{G}$

$E(\bar{G}) = G_{full} = -1$

$Var(\bar{G}) = E[\bar{G}^2] - E^2[\bar{G}]$

$E[\bar{G}] = 1$

PMF $\bar{G} \to \{-4, -3, -2, -1, 0, 1, 2\}$

$E[\bar{G}^2] = \frac{7}{2}$

$Var(\bar{G}) = \frac{7}{2} - 1 = \frac{5}{2}$

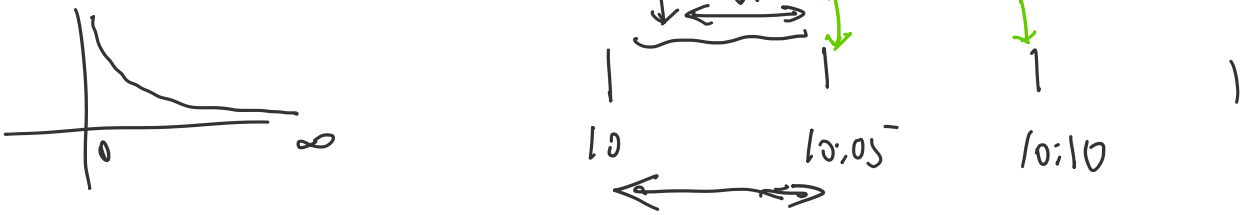
	E	Var
1 Random sample	g_{full}	5
2 Random samples	g_{full}	$\frac{5}{2}$

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$$\text{Var}(X) = \int_0^\infty \left(x - \frac{1}{\lambda}\right)^2 \cdot \lambda e^{-\lambda x}\, dx$$

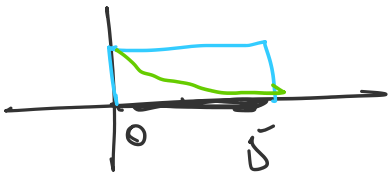
$X \sim Exp(\lambda)$

$$E(X^2) - \mu_X^2$$



$$A \sim U[10,\ 10:05]$$

W



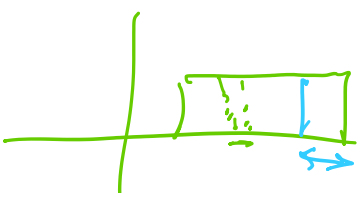
X , Prob. dist. , Var(X) → fn of the PMF

$$E(X^2) - \mu_X^2$$

$X \sim Unif.(a, b)$

$$\mu_X = \frac{a+b}{2}, \quad E(X^2) = \frac{b^3 - a^3}{3(b-a)}$$

$$\text{Var}(X) = \frac{(b-a)^2}{12}$$



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$X \sim PMF\ p_X(k) \rightarrow$ 0, 1, 2 ; $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$

$[E(X)]^2$

$E[X^2] = \Sigma$

$0^2 \times \frac{1}{4} + 1^2 \times \frac{1}{2} + 2^2 \times \frac{1}{4}$

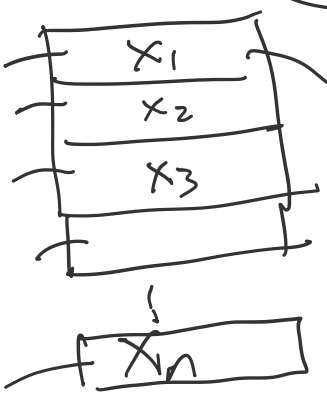
$= \frac{1}{4} \quad = \frac{2}{4} \quad = \frac{4}{4}$

Time avg. vs Statistical Avg

Empirical

$X_i \sim f_X$

Empirical var. vs Statistical var $\quad$ var(f_X)



$\bar{\mu}_X = \frac{x_1 + x_2 + \cdots + x_n}{n}$

$\frac{1}{n} \sum_{i=1}^{n} (x_i - \bar{\mu}_X)^2$

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